

Subject: New Portfolio Project: Delivery Operations & Delay Analysis (Python + Tableau)

Hi everyone,

It's time to put your **Pandas**, **NumPy**, and **Tableau** skills into action with a real-world portfolio project that you will publish to your GitHub.

The Scenario: Last-Mile Delivery Performance Analysis

Your Role: Operations Analyst at an on-demand delivery platform.

The Business Problem: Delivery times have surged over the past quarter, driving up customer complaints. Management needs you to find the bottleneck: *Is it kitchen prep delays, traffic/transit time, or peak hours?*

Your Tasks:

1. Data Prep (Python - Pandas & NumPy):

- Clean raw timestamps and handle missing/invalid values.
- Remove outliers.
- Feature engineering: Calculate `Prep Time`, `Transit Time`, and `Total Delivery Time`.
- Create categories for time-of-day (e.g., *Lunch Peak*, *Dinner Rush*) and flag delayed orders.
- Export the clean dataset to `.csv`.

2. Dashboard (Tableau):

- Build an interactive dashboard showing delay hotspots, peak-hour bottlenecks, and vehicle performance.
- Publish it to **Tableau Public**.

3. GitHub Portfolio Requirements:

- Create a clean repository with your Jupyter Notebook (`.ipynb`).
- Write a professional `README.md` that includes:
 - A 2-sentence summary of the business problem.
 - 2–3 key insights/findings.
 - A screenshot and a live link to your Tableau Public dashboard.

 **Dataset Link:** <https://www.kaggle.com/datasets/gauravmalik26/food-delivery-dataset?select=train.csv>

Treat this like a real task from a manager—this will be a standout piece on your GitHub for future recruiters.

Happy coding!

Best regards,
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